~~Long~~ Short division

MAGIC NUMBER TRICK

There are several tricks that tell you whether a number is divisible by 3, 4, 5, 9, 10, 11. ÷

* To find out if a NUMBER is divisible by 3, add up the digits. If they add up to a MULTIPLE of 3, the number is divisible by 3. For instance, 192 must be divisible by 3 because 1 + 9 + 2 = 12.

* A number is DIVISIBLE by 4 if the last two digits are 00 or a multiple of 4.

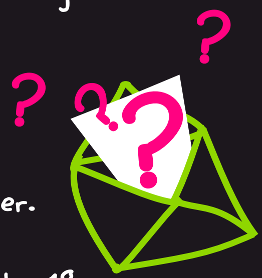
* A number is divisible by 5 if the LAST DIGIT is 5 or 0.

* A number is divisible by 9 if all the digits add up to a MULTIPLE of 9. For instance, 201,915 must be divisible by 9 because 2 + 0 + 1 + 9 + 1 + 5 = 18.

* A numbers is DIVISIBLE by 10 IF THE LAST DIGIT is 0.

* To FIND OUT IF A NUMBER IS DIVISIBLE BY 11, START WITH THE DIGIT ON THE LEFT, subtract the next digit from it, add the next, subtract the next, and so on. IF THE ANSWER is 0 or 11, then the original number is divisible by 11. For instance, is 35706 divisible by 11? 3 − 5 + 7 − 0 + 6 = 11, so the answer is yes.

1. First write down the number nine (as a word) on a piece of paper and seal it in an envelope.
2. Give the envelope to a friend and tell them to keep it safe.
3. Next give your friend a calculator. Ask them to key in the last two numbers of their phone number.
4. Add the number of pounds in their pocket.
5. Add their age.
6. Add the number of their house.
7. Subtract the number of brothers and sisters they have.
8. Subtract 12.
9. Ask them to add their favourite number.



10. Multiply the answer by 18.
11. Ask them to add up all the digits in the answer.
12. If the answer is more than 1 digit long, ask them to add up the digits again. Carry on until there's only one digit, which will be 9.
13. Finally, tell your friend to open the envelope and read out the number.